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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

II Semester B. E. Regular / Supplementary Examinations Aug-2024

VECTOR CALCULUS, LAPLACE TRANSFORM AND NUMERICAL METHODS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of mathematics Handbook is permitted. Do not write anything on handbook.

PART-A

			M	BT	CO
1	1.1	If vector $\vec{f} = 2ax\hat{\imath} - ay\hat{\jmath} + 4z\hat{k}$ is solenoidal, then the value of the constant 'a' is _____.	01	1	1
	1.2	The normal vector to the surface $y^2 - 4x^2 + 3z = 3$ is _____.	01	1	1
	1.3	The region of convergence for the Laplace transform of $sin(2t)$ is _____.	01	1	1
	1.4	If $L[f(t)] = \frac{5}{s(s^2+7),}$ then $L[f(2t)] =$ _____.	01	1	1
	1.5	$L^{-1}\left[\frac{1}{(s+2)^5}\right] =$ _____.	02	1	2
	1.6	$L[\sqrt{t}\, e^{-\frac{t}{2}}] =$ _____.	02	2	1
	1.7	Inverse Laplace transform of $\frac{s-3}{s^2-6s+2}$ is _____.	02	2	1
	1.8	Given $\frac{dy}{dx} = x + y^2, y(0) = 1, h = 0.1, k_1 = 0.1, k_2 = 0.1152, k_3 = 0.1168$, *then* the solution of the differential equation at $x = 0.1$ using Runge kutta method of fourth order is _____.	02	2	2
	1.9	First approximate value of a real root of the equation $2x - \log_{10} x = 7$ in the interval [3.5,4] by false position method is _____.	02	1	1
	1.10	If $\vec{F} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ and V is the region bounded by cube $x = 0, y = 0, z = 0, x = 1, y = 2, z = 3$, then $\iiint_v \nabla.\vec{F}$ is _____.	02	2	2
	1.11	If $\vec{F} = (x^2 + 5y^2)\hat{\imath} + (3y^2 - 4x)\hat{\jmath}$, then evaluate $\int_c \vec{F}.d\vec{r}$ along the straight line from (0,0) to (0,2).	02	2	2
	1.12	Compute gradient of the scalar field $\phi = r + zcos\theta$, specified in the cylindrical coordinates (r,θ,z).	02	2	2

PART-B

2	a	Find the directional derivative of $\phi(x,y,z) = \frac{1}{x^2+y^2+z^2}$ at the point (1,2,−3) in the direction of $\hat{\imath} + 3\hat{\jmath} + \hat{k}$.	06	2	2
	b	Show that the vector $\vec{F} = (y^2\cos(x) + z^3)\hat{\imath} + (2ysin(x) - 4)\hat{\jmath} + 3xz^2\hat{k}$ is irrotational and find its scalar potential such that $\vec{F} = \nabla\phi$	06	3	3
	c	If $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ *and* $r = \lvert\vec{r}\rvert$, then show that $\nabla r^n = nr^{n-2}\vec{r}$	04	2	2
3	a	Using divergence theorem, evaluate $\iint_s [(x^2 - yz)\,\hat{\imath} + (y^2 - zx)\hat{\jmath} + (z^2 - xy)\hat{k}].\hat{n}ds$, over the surface of the rectangular parallelepiped $0 \le x \le a, 0 \le y \le b, 0 \le z \le c$	08	2	3
	b	Using Green's theorem in the plane evaluate $\int_c \{(3x^2 - y^2)dx + (x^2 + 5y^2)dy\}$, C is the boundary of the region bounded by $x = 0, y = 0, x + y = 2$.	08	2	3

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		OR			
4	a	Using Stokes theorem evaluate $\int_c (x+y)dx + (2x-z)dy + (y+z)dz$ *where* C is the boundary of the triangle with vertices (2,0,0), (0,3,0) and (0,0,6).	10	3	3
	b	Evaluate $\int_c \vec{F}.d\vec{r}$ *where* $\vec{F} = (xy)\hat{i} + (x^2+y^2)\hat{j}$ along			
		i) The path of the straight line from (0,0) to (1,0) and then to (1,1).	06	2	2
		ii) The straight line joining the origin and (1,2).			
5	a	Find the Laplace transform of $\int_0^t \frac{\sin t}{t}\,dt + te^{-t}\cos^2 2t$	08	2	2
	b	Express the piecewise continuous function $f(t) = \begin{cases} 1, & 0 < t \le 1 \\ t, & 1 < t \le 2 \\ t^2, & t > 2 \end{cases}$ in terms of the unit step function and also find its Laplace transform.	08	2	3
		OR			
6	a	i) Evaluate $\int_0^\infty e^{-6t} t \sin(t)\,dt$ *using* Laplace transform.			
		ii) Find the Laplace transform of the function $(e^{-at} - \cos(bt))/t$	08	2	2
	b	Show that the Laplace transform of the periodic function $f(t)$ is $\frac{a}{s}\tanh\left(\frac{as}{2}\right)$, *where* $f(t) = \begin{cases} k, 0 \le t \le a \\ -k, a \le t < 2a \end{cases}$ $f(t+2a) = f(t)$. Also sketch the waveform for $f(t)$.	08	2	2
7	a	Find the inverse Laplace transform of $\frac{4s+5}{(s+1)^2(s+2)}$	08	2	2
	b	Using convolution theorem, obtain the inverse Laplace transform of $\frac{1}{(s+1)(s^2+4)}$	08	3	3
		OR			
8	a	Solve $\frac{d^2y}{dt^2} + \frac{dy}{dt} = t^2 + 2t, y(0) = 4, y'(0) = -2$ using Laplace transform.	08	3	4
	b	Find			
		i) $L^{-1}\left[\cot^{-1}\left(\frac{2}{s+1}\right)\right]$			
		ii) $L^{-1}\left[e^{-4s}\left(\frac{s+3}{s^2+4s+13}\right)\right]$	08	2	2
9	a	Use the Runge Kutta method of fourth order with $h = 0.1$ to find approximate value of the solution of the initial value problem $\frac{dy}{dx} = 2x^2 + 3y^2 - 2, y(2) = 1$ *at* $x = 2.1$.	08	2	4
	b	Employ Taylor series method to obtain the approximate value of y at $x = 0.1$ for the differential equation $\frac{dy}{dx} = e^x - y^2; y(0) = 1$ considering the terms upto fourth degree.	08	2	2
		OR			
10	a	Apply Milne's predictor-corrector method to find the solution of the differential equation $\frac{dy}{dx} + x^3 = y$ at $x = 0.4$ given that $y(0) = 1$, $y(0.1) = 1.1051$, $y(0.2) = 1.2210$, $y(0.3) = 1.3477$	10	2	2
	b	Using Newton Raphson method, find the root of the equation $x\cos x = x^2$ correct to four decimal places choosing the initial guess $x_0 = 1$.	06	2	2

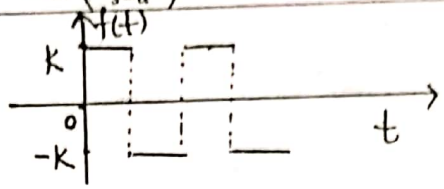
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	PART-A	
6		1
1.1	$a = -4$	1
1.2	$\nabla\phi = -8x\hat{i} + 2y\hat{j} + 3\hat{k}$	1
1.3	$s > 0$	1
1.4	$\frac{20}{s(s^2+28)}$	2
1.5	$\frac{24e^{-2t}}{t^4} = \frac{t^4}{24}e^{-2t}$	1+1
1.6	$\frac{\sqrt{\pi}}{2\left(s+\frac{1}{2}\right)^{\frac{3}{2}}}$	2
1.7	$e^{3t}\cosh(\sqrt{7}t)$	1
1.8	$k_4 = 0.1347$ $y(0.1) = 1.1164$	1 2
1.9	$x_1 = 3.7887$, $f(a) = \quad f(b) =$	1+1
1.10	$\iiint_V \nabla.\vec{F}\, dV = \int_0^1\int_0^2\int_0^3 3\, dV = 18$	2
1.11	$\int_C \vec{F}.d\vec{r} = \int_0^2 3y^2 dy = 8$	1+1
1.12	$\nabla\phi = \frac{\partial\phi}{\partial r}\hat{e_r} + \frac{\partial\phi}{\partial\theta}\hat{e_\theta} + \frac{\partial\phi}{\partial z}\hat{e_z}$, $\nabla\phi = \hat{e_r} - z\sin\theta\hat{e_\theta} + \cos\theta\hat{e_z}$	
	PART-B	
2a	$\nabla\phi = -\frac{2x}{(x^2+y^2+z^2)^2}\hat{i} - \frac{2y}{(x^2+y^2+z^2)^2}\hat{j} - \frac{2z}{(x^2+y^2+z^2)^2}\hat{k}$ $\nabla\phi_{(1,2,-3)} = \frac{-2\hat{i}-4\hat{j}+6\hat{k}}{196}$ Directional derivative of $\phi = \nabla\phi.\hat{c} = \frac{-2\hat{i}-4\hat{j}+6\hat{k}}{196}.\frac{\hat{i}+3\hat{j}+\hat{k}}{\sqrt{11}} = = \frac{-2-12+6}{196\sqrt{11}} = -\frac{8}{196\sqrt{11}} = \frac{-2}{49\sqrt{11}}$	2 1 1+1
2b	$\text{curl}\,\vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2\cos(x)+z^3 & 2y\sin(x)-4 & 3xz^2 \end{vmatrix}$ $\text{curl}\,\vec{F} = (0-0)\hat{i} - (3z^2-3z^2)\hat{j} + (2y\cos(x)-2y\cos(x))\hat{k} = \vec{0}$ $\vec{F} = (y^2\cos(x)+z^3)\hat{i} + (2y\sin(x)-4)\hat{j} + 3xz^2\hat{k}$ $\frac{\partial\phi}{\partial x} = y^2\cos(x)+z^3$, $\frac{\partial\phi}{\partial y} = 2y\sin(x)-4$, $\frac{\partial\phi}{\partial z} = 3xz^2$ $d\phi = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz$, $\phi = y^2\sin(x) + xz^3 - 4y + c$	2 1+3
2c	$r^2 = x^2+y^2+z^2$ and $\frac{\partial r}{\partial x} = \frac{x}{r}, \frac{\partial r}{\partial y} = \frac{y}{r}, \frac{\partial r}{\partial z} = \frac{z}{r}$ $\nabla r^n = nr^{n-1}\frac{x}{r}\hat{i} + nr^{n-1}\frac{y}{r}\hat{j} + nr^{n-1}\frac{z}{r}\hat{k} = nr^{n-2}\vec{r}$	1 2+1
3a	$\int\int_S \vec{F}.\hat{n}\, dS = \int\int\int_V div\,\vec{F}\, dV = \int_{x=0}^{a}\int_{y=0}^{b}\int_{z=0}^{c} 2(x+y+z)\, dz\, dy\, dx$	

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	$\iiint_V div\,\vec{F}\,dV = 2\int_0^a\int_0^b\left[cx+cy+\frac{c^2}{2}\right]dy\,dx = 2\int_0^a\left(cxb+\frac{cb^2}{2}+\frac{c^2}{2}b\right)dx$ $= 2\left(\frac{cba^2}{2}+\frac{cb^2}{2}a+\frac{c^2}{2}ba\right), \therefore \iiint_V div\,\vec{F}\;dV = abc(a+b+c)$.	2+2+2
3b	$\oint_C Pdx+Qdy = \iint_R\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dx\,dy,\ P=3x^2-y^2 \Rightarrow \frac{\partial P}{\partial y}=-2y$ and $Q=x^2+5y^2 \Rightarrow \frac{\partial Q}{\partial x}=2x$ $\iint_R\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dx\,dy = \int_{y=0}^{2}\int_{x=0}^{2-y}(2x+2y)dxdy = 2\int_0^2\left[\frac{x^2}{2}+xy\right]_0^{2-y}dy$ $= 2\int_0^2\left(\frac{(2-y)^2}{2}+(2-y)y\right)dy = 2\int_0^2\left(\frac{1}{2}(4+y^2-4y)+(2y-y^2)\right)dy$ $= 2\left[\frac{1}{2}\left(4y+\frac{y^3}{3}-4\frac{y^2}{2}\right)+\left(2\frac{y^2}{2}-\frac{y^3}{3}\right)\right]_0^2 = \frac{16}{3}$	Graph-1 1 2+2 2 2
4a	By Stokes theorem we have $\oint_C \vec{F}.d\vec{r} = \iint_S curl\,\vec{F}.\hat{n}\,ds$ $\nabla\times\vec{F} = \begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\ \frac{\partial}{\partial x}&\frac{\partial}{\partial y}&\frac{\partial}{\partial z}\\ x+y&2x-z&y+z\end{vmatrix} = 2\hat{i}+\hat{k}$ $\nabla\phi = 3\hat{i}+2\hat{j}+\hat{k}\ , \lvert\nabla\phi\rvert = \sqrt{14}, \hat{n} = \frac{x\hat{i}+y\hat{j}+z\hat{k}}{a},\ curl\vec{F}.\hat{n} = \frac{7}{\sqrt{14}}$ $\iint_S curl\,\vec{F}.\hat{n}\,ds = \frac{7}{\sqrt{14}}\iint_R\frac{dx\,dy}{\frac{1}{\sqrt{14}}} = 7(Area\ of\ the\ triangle) = 7\times\frac{1}{2}\times2\times3 = 21.$	Graph1 2 2+2 2+1
4b	$\int_C \vec{F}.d\vec{r} = \int_{OA}\vec{F}.d\vec{r}+\int_{AB}\vec{F}.d\vec{r}$ Along OA: $y=0$ and $dy=0$, $\int_{OA}\vec{F}.d\vec{r}=0$ Along AB: $x=1$ and $dx=0$, $\int_{AB}\vec{F}.d\vec{r} = \int_0^1(1+y^2)\,dy = \frac{4}{3}$, $\int_C\vec{F}.d\vec{r} = \frac{4}{3}$ ii) Straight line equation is $y=2x$ and $dy=2dx$ $\int_C\vec{F}.d\vec{r} = \int_{x=0}^{1}x(2x)\,dx+(x^2+4x^2)2dx = \int_0^1 12x^2dx = 4$	~~Graph~~1 1 1+1 2+1 (3+3)
5a	$L\left[\int_0^t\frac{\sin t}{t}dt\right] = \int_s^\infty\frac{1}{s^2+1}ds = \frac{\cot^{-1}(s)}{s}$ $L[\cos^2(2t)] = L\left[\frac{1+\cos(4t)}{2}\right] = \frac{1}{2}\left[\frac{1}{s}+\frac{s}{s^2+16}\right]$, $L[t\cos^2(2t)] = \frac{1}{2}\left[\frac{1}{s^2}+\frac{s^2-16}{(s^2+16)^2}\right]$ $L[te^{-t}\cos^2(2t)] = \frac{1}{2}\left[\frac{1}{(s+1)^2}+\frac{(s+1)^2-16}{((s+1)^2+16)^2}\right]$ $L\left[\int_0^t\frac{\sin t}{t}dt+te^{-t}\cos^2 2t\right] = \frac{\cot^{-1}(s)}{s}+\frac{1}{2}\left[\frac{1}{(s+1)^2}+\frac{(s+1)^2-16}{((s+1)^2+16)^2}\right]$	1+2+1 1+2+1 3
5b	$f(t) = 1+(t-1)u(t-1)+(t^2-t)u(t-2)$ $L[f(t)] = L[1]+L[(t-1)u(t-1)]+L[(t^2-t)u(t-2)]$ $L[f(t)] = L[1]+e^{-s}L[t]+e^{-2s}L[t^2+3t+2] = \frac{1}{s}+\frac{e^{-s}}{s^2}+e^{-2s}\left[\frac{2}{s^3}+\frac{3}{s^2}+\frac{2}{s}\right]$	1+1+3
6a	$i) L(\sin t) = \frac{1}{s^2+1}, L(t\sin t) = -\frac{d}{ds}\left(\frac{1}{s^2+1}\right) = \frac{2s}{(s^2+1)^2}, = \frac{12}{1369}$	2+2

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	$ii) L[e^{at} - \cos(bt)] = \frac{1}{s-a} - \frac{s}{s^2+b^2}$, $L\left[\frac{e^{at}-\cos(bt)}{t}\right] = \int_s^\infty \left[\frac{1}{s-a} - \frac{s}{s^2+b^2}\right] ds = \log_e\left(\frac{s-a}{\sqrt{s^2+b^2}}\right)\Big]_s^\infty = \log_e\left(\frac{\sqrt{s^2+b^2}}{s-a}\right)$	1+1 2
6b	$L[f(t)] = \frac{1}{1-e^{-sT}}\int_0^T e^{-st} f(t)\,dt$ $= \frac{1}{1-e^{-2as}}\left[\int_0^a e^{-st}(k)dt + \int_a^{2a} e^{-st}(-k)dt\right]$ $= \frac{1}{1-e^{-2as}} \times \frac{k}{s}[1 - 2e^{-as} + e^{-2as}]$ $= \frac{1}{1-e^{-2as}} \times \frac{k}{s}[(1-e^{-as})^2] = \frac{k}{s}\tanh\left(\frac{as}{2}\right)$ 	Graph 2 1+1
7a	$L^{-1}\left[\frac{4s+5}{(s+1)^2(s+2)}\right] = L^{-1}\left[\frac{3}{s+1} + \frac{1}{(s+1)^2} - \frac{3}{s+2}\right] = 3e^{-t} + te^{-t} - 3e^{-2t}$	1+3 1+2+1
7b	$F(s) = \frac{1}{s+1}$, $G(s) = \frac{1}{s^2+4}$, $f(t) = e^{-t}$, $g(t) = \frac{1}{2}\sin 2t$ $L^{-1}\left[\frac{1}{((s+1)(s^2+9))}\right] = \frac{1}{2}\int_0^t e^{-u}\sin(2t-2u)du$ $= \frac{1}{2}sin2t\int_0^t e^{-u}cos2udu - \frac{1}{2}sin2t\int_0^t e^{-u}sin2udu$ $= \frac{1}{2}sin2t\left[\frac{e^{-t}(-cos2t+2sin2t)}{5} + \frac{1}{5}\right] - \frac{1}{2}cos2t\left[\frac{e^{-t}(-sin2t-2cos2t)}{5} + \frac{2}{5}\right]$ $= \frac{e^{-t}}{5} + \frac{1}{10}sin2t - \frac{1}{5}cos2t$	2 2+2
8a	$s^2Y(s) - sy(0) - y'(0) + sY(s) - y(0) = \frac{2}{s^3} + \frac{2}{s^2}$ $s(s+1)Y(s) = 4s + 2 + \frac{2(s+1)}{s^3}$, $Y(s) = \frac{4}{s+1} + \frac{2}{s(s+1)} + \frac{2}{s^4} = \frac{4}{s+1} + \frac{2}{s} - \frac{2}{s+1} + \frac{2}{s^4}$ $y(t) = 4e^{-t} + 2 - 2e^{-t} + \frac{t^3}{3} = 2e^{-t} + 2 + \frac{t^3}{3}$	2 2 2+2
8b	(i) $F(s) = \cot^{-1}\left(\frac{2}{s+1}\right) \Rightarrow F'(s) = \frac{2}{(s+1)^2+4} \Rightarrow L^{-1}[F'(s)] = e^{-t}\sin(2t)$ $L^{-1}[F(s)] = -\frac{1}{t}L^{-1}[F'(s)] = L^{-1}\left[\cot^{-1}\left(\frac{2}{s+1}\right)\right] = -\frac{e^{-t}\sin(2t)}{t}$ (ii) $L^{-1}\left[\frac{s+3}{s^2+4s+13}\right] = L^{-1}\left[\frac{s+2+1}{(s+2)^2+3^2}\right] = L^{-1}\left[\frac{s+2}{(s+2)^2+3^2} + \frac{1}{(s+2)^2+3^2}\right]$ $L^{-1}\left[\frac{s+3}{s^2+4s+13}\right] = e^{-2t}\cos(3t) + \frac{1}{3}e^{-2t}\sin(3t)$ $L^{-1}\left[e^{-4s}\left(\frac{s+3}{s^2+4s+13}\right)\right] = e^{-2(t-4)}\left[\cos(3t-12) + \frac{\sin(3t-12)}{3}\right]u(t-4)$	2 2 1 2 1
9a	$k_1 = 0.9$, $k_2 = 1.2712$, $k_3 = 1.4431$, $k_4 = 2.4726$ $k = \frac{k_1+2k_2+2k_3+k_4}{6} = 1.4669$, $y_1 = y_0 + k = 2.4669$	1+1+2+2 1+1
9b	$y' = e^x - y^2 \Rightarrow y_0' = 0, y'' = e^x - 2yy' \Rightarrow y_0'' = 1$. $y''' = e^x - 2(yy'' + (y')^2) \Rightarrow y_0''' = -1, y^{iv} = e^x - 2(yy''' + 3y'y'') \Rightarrow y_0^{iv} = 3$ $y(0.1) = 1.0048$	1+2+1+1 1+1
10a	$y(0) = 1$, $y(0.1) = 1.1051$, $y(0.2) = 1.2210$, $y(0.3) = 1.3477$ $f_1 = 1.1041$, $f_2 = 1.213$, $f_3 = 1.3207$ Milne's Predictor formula $y_{(4)}^{(p)} = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3) = 1 + \frac{0.4}{3}(2(1.1041) - 1.213 + 2(1.3207)) = 1.4849$ $f_4 = 1.4209$ Milne's Corrector formula	3 3 1

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	$y_4^{(c)} = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4) = 1.2210 + \frac{0.1}{3}(1.213 + 4(1.3207) + 1.4209) = 1.4849$	2
10b	$f(x) = x\cos x - x^2$, $f'(x) = \cos x - x\sin x - 2x$ Formula $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, $x_1 = 0.8002, x_2 = 0.7440, x_3 = 0.7391, x_4 = 0.7390$	1+1+1+1
Appropriate marks may be awarded for the alternate methods.		

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